

Notebook 2- Feature Relationship Analysis

Section-1 Objective

This notebook investigates the statistical relationships between neighbourhood-level socioeconomic indicators and low-income prevalence within the City of Toronto.

The objective is to quantify the strength and direction of the relationships between the selected independent variables and the dependent variable, **Low_Income_AfterTax_Pct**, prior to predictive modelling.

The analysis combines descriptive statistics, Pearson and Spearman correlation analysis, graphical exploration, and multicollinearity assessment to identify the socioeconomic indicators most strongly associated with neighbourhood vulnerability.

The findings from this notebook provide the statistical evidence required to answer **Research Question 1** and guide feature selection for the predictive modelling stage.

Section 2. Project Input & Output

Input

Dataset

- Machine learning-ready dataset produced in **Notebook 1**
- File: `neighbourhood_profiles_2016_ml_ready.csv`

The dataset contains standardized socioeconomic indicators for the City of Toronto and its 140 neighbourhoods. Each row represents one neighbourhood, while each column represents either a socioeconomic predictor variable or the target variable, **Low_Income_AfterTax_Pct**.

Output

This notebook will produce:

- Descriptive statistical summary of all selected variables
- Pearson correlation analysis
- Spearman rank correlation analysis
- Correlation heatmap
- Ranked socioeconomic predictors based on their association with low-income prevalence
- Scatterplots of the strongest relationships

- Variance Inflation Factor (VIF) analysis for multicollinearity assessment
- Interpretation of statistically significant findings
- Evidence-based answer to **Research Question 1**

Dataset

- File: 'neighbourhood_profiles_2016_analysis_ready.csv'
'02_Statistical_Analysis_Results.xlsx'

In [112...

```
import pandas as pd
import numpy as np

import matplotlib.pyplot as plt
import seaborn as sns

from pathlib import Path

from scipy.stats import pearsonr
from scipy.stats import spearmanr

pd.set_option("display.max_columns", None)
pd.set_option("display.float_format", "{:.2f}".format)
```

In [113...

```
from pathlib import Path

project_root = Path.cwd().parents[1]

print(project_root)
```

c:\Users\ksalyamova\Desktop\CIND 820\CIND820-Capstone-Project

Section-3 Load Machine Learning Dataset

In [114...

```
input_path = (project_root / "data" / "processed" / "neighbourhood_profiles_2016_ml_re

df = pd.read_csv(input_path)

print("=" * 60)
print("DATASET LOADED")
print("=" * 60)

print(f"Dataset loaded from:\n{input_path}")
```

=====

DATASET LOADED

=====

Dataset loaded from:

c:\Users\ksalyamova\Desktop\CIND 820\CIND820-Capstone-Project\data\processed\neighbo
urhood_profiles_2016_ml_ready.csv

Section 4- Dataset Validation

Before conducting statistical analyses, the machine learning-ready dataset is inspected to verify its structure, dimensions, variable names, data types, and overall completeness.

This verification step confirms that the exported dataset from Notebook 1 has been loaded correctly and is suitable for subsequent statistical analyses. It also provides an opportunity to identify any unexpected structural issues prior to correlation analysis.

In [115...

#Dataset Dimensions

```
print("=" * 60)
print("DATASET DIMENSIONS")
print("=" * 60)

print(f"Rows: {df.shape[0]}")
print(f"Columns: {df.shape[1]}")
```

```
=====
DATASET DIMENSIONS
=====
Rows: 140
Columns: 14
```

In [116...

Dataset Preview

display(df.head())

	Neighbourhood	Population	Participation_Rate	Employment_Rate	Unemployment_Rate	
0	Agincourt North	29113	55.40	50.00	9.80	
1	Agincourt South-Malvern West	23757	59.00	53.20	9.80	
2	Alderwood	12054	66.50	62.40	6.10	
3	Annex	30526	70.60	65.80	6.70	
4	Banbury-Don Mills	27695	59.90	55.60	7.20	



In [117...

Data Types

```
print("=" * 60)
print("DATA TYPES")
print("=" * 60)

display(df.dtypes)
```

```
=====
DATA TYPES
=====
```

```

Neighbourhood      object
Population         int64
Participation_Rate float64
Employment_Rate    float64
Unemployment_Rate  float64
Bachelors_Degree   int64
No_Diploma          int64
Immigrant_Population int64
Visible_Minority    int64
Renters            int64
Core_Housing_Need   int64
Unaffordable_Housing float64
Lone_Parent_Families int64
Low_Income_AfterTax_Pct float64
dtype: object

```

In [118...

```

# Missing Values

print("=" * 60)
print("MISSING VALUES")
print("=" * 60)

display(df.isna().sum())

```

```

=====
MISSING VALUES
=====

```

```

Neighbourhood      0
Population         0
Participation_Rate  0
Employment_Rate    0
Unemployment_Rate  0
Bachelors_Degree   0
No_Diploma          0
Immigrant_Population 0
Visible_Minority    0
Renters            0
Core_Housing_Need   0
Unaffordable_Housing 0
Lone_Parent_Families 0
Low_Income_AfterTax_Pct 0
dtype: int64

```

In [119...

```

# Dataset Information

df.info()

```

```

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 140 entries, 0 to 139
Data columns (total 14 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   Neighbourhood                        140 non-null    object
1   Population                          140 non-null    int64
2   Participation_Rate                  140 non-null    float64
3   Employment_Rate                     140 non-null    float64
4   Unemployment_Rate                   140 non-null    float64
5   Bachelors_Degree                    140 non-null    int64
6   No_Diploma                          140 non-null    int64
7   Immigrant_Population                140 non-null    int64
8   Visible_Minority                    140 non-null    int64
9   Renters                             140 non-null    int64
10  Core_Housing_Need                   140 non-null    int64
11  Unaffordable_Housing                140 non-null    float64
12  Lone_Parent_Families                140 non-null    int64
13  Low_Income_AfterTax_Pct             140 non-null    float64
dtypes: float64(5), int64(8), object(1)
memory usage: 15.4+ KB

```

Interpretation

The inspection confirms that the machine learning-ready dataset was successfully loaded and retains the structure established in Notebook 1.

The dataset contains one observation for each of the 140 neighbourhoods, with standardized socioeconomic predictor variables and a single dependent variable representing neighbourhood-level low-income prevalence.

All variables exhibit the expected data types, and no missing values were detected. These results confirm that the dataset is complete and suitable for the statistical analyses presented in the following sections.

Section-7. Population-Based Ratio Feature Engineering

Several socioeconomic indicators in the machine learning-ready dataset are recorded as absolute population counts rather than proportions. Directly comparing these count-based variables across neighbourhoods of different population sizes may produce misleading statistical relationships driven primarily by neighbourhood size rather than underlying socioeconomic conditions.

To improve comparability, selected count-based independent variables are transformed into population-based percentages by dividing each count by the corresponding neighbourhood population and multiplying by 100. This feature engineering step standardizes the selected variables while preserving their substantive interpretation.

The resulting percentage-based variables provide a more appropriate basis for subsequent correlation analysis, multicollinearity assessment, dimensionality reduction, clustering, and predictive modelling.

In [120...

```
count_variables = [
    "Bachelors_Degree",
    "No_Diploma",
    "Immigrant_Population",
    "Visible_Minority",
    "Renters",
    "Lone_Parent_Families"]

for column in count_variables:
    df[column + "_Pct"] = (df[column] / df["Population"]) * 100

print("=" * 60)
print("FEATURE NORMALIZATION COMPLETE")
print("=" * 60)

display(
    df[
        "Neighbourhood",
        "Population",
        "Bachelors_Degree",
        "Bachelors_Degree_Pct",
        "Immigrant_Population",
        "Immigrant_Population_Pct"
    ].head())
```

```
=====
FEATURE NORMALIZATION COMPLETE
=====
```

	Neighbourhood	Population	Bachelors_Degree	Bachelors_Degree_Pct	Immigrant_Population_Pct
0	Agincourt North	29113	4380	15.04	1995
1	Agincourt South-Malvern West	23757	4210	17.72	1505
2	Alderwood	12054	1660	13.77	397
3	Annex	30526	9135	29.93	827
4	Banbury-Don Mills	27695	6500	23.47	1315

In [121...

```
# Replace count-based variables with percentage-based variables

df = df.drop(columns=[
    "Bachelors_Degree",
    "No_Diploma",
    "Immigrant_Population",
    "Visible_Minority",
```

```
"Renters",
"Lone_Parent_Families"]])

print(df.columns)
```

```
Index(['Neighbourhood', 'Population', 'Participation_Rate', 'Employment_Rate',
      'Unemployment_Rate', 'Core_Housing_Need', 'Unaffordable_Housing',
      'Low_Income_AfterTax_Pct', 'Bachelors_Degree_Pct', 'No_Diploma_Pct',
      'Immigrant_Population_Pct', 'Visible_Minority_Pct', 'Renters_Pct',
      'Lone_Parent_Families_Pct'],
      dtype='object')
```

Interpretation

The selected count-based independent variables were successfully transformed into population-based percentages, reducing the influence of differences in neighbourhood population size while improving comparability across Toronto neighbourhoods.

This feature engineering step ensures that the subsequent statistical analyses examine socioeconomic characteristics relative to neighbourhood population rather than absolute counts. The resulting percentage-based variables provide a more meaningful basis for correlation analysis, dimensionality reduction, clustering, and predictive modelling conducted in the subsequent notebooks.

Section-8. Descriptive Statistics

Before quantifying the relationships between socioeconomic indicators and neighbourhood-level low-income prevalence, descriptive statistics are examined to summarize the characteristics of each variable.

This preliminary analysis provides measures of central tendency, variability, and distribution, allowing an initial assessment of the socioeconomic conditions represented within the dataset.

The descriptive statistics also provide context for the subsequent Pearson and Spearman correlation analyses by identifying the range and variability of each predictor and the target variable.

In [122...

```
print("=" * 60)
print("DESCRIPTIVE STATISTICS")
print("=" * 60)

display(df.describe().T.round(2))
```

```
=====
DESCRIPTIVE STATISTICS
=====
```

	count	mean	std	min	25%	50%	75%
Population	140.00	19511.22	10033.59	6577.00	12019.50	16749.50	23854.50
Participation_Rate	140.00	64.55	6.11	52.20	59.88	63.60	69.12
Employment_Rate	140.00	59.27	6.52	47.30	54.68	58.55	64.30
Unemployment_Rate	140.00	8.30	1.90	4.50	6.90	8.20	9.62
Core_Housing_Need	140.00	1720.14	1002.88	115.00	967.50	1432.50	2287.50
Unaffordable_Housing	140.00	35.07	6.94	17.50	31.08	35.60	38.52
Low_Income_AfterTax_Pct	140.00	19.51	7.89	4.50	14.10	18.55	23.95
Bachelors_Degree_Pct	140.00	19.21	7.16	5.28	13.93	19.29	24.38
No_Diploma_Pct	140.00	20.33	3.90	11.95	17.42	20.98	23.53
Immigrant_Population_Pct	140.00	44.22	13.50	19.59	31.64	45.36	55.46
Visible_Minority_Pct	140.00	46.48	21.96	11.86	27.24	42.52	66.08
Renters_Pct	140.00	18.97	8.97	2.10	13.56	17.65	23.75
Lone_Parent_Families_Pct	140.00	5.59	1.85	2.43	4.17	5.46	6.53

Interpretation

The descriptive statistics summarize the socioeconomic characteristics of the 140 Toronto neighbourhoods included in the machine learning-ready dataset. All variables contain 140 complete observations, confirming that the dataset is suitable for subsequent statistical analysis.

The independent variables exhibit substantial variation across neighbourhoods, reflecting differences in population, education, employment, housing, and demographic composition. Converting count-based variables into population-normalized percentages improves comparability by reducing the influence of neighbourhood size.

The dependent variable, Low_Income_AfterTax_Pct (LIM-AT), ranges from 4.50% to 45.50%, with a mean of 19.51% and a median of 18.55%. This indicates that, on average, approximately one in five residents falls below the Low Income Measure after tax, while the wide range highlights considerable socioeconomic differences between Toronto neighbourhoods.

Overall, these findings confirm sufficient variability for subsequent correlation analysis, feature selection, and predictive modelling.

In [123...

```
# Create DataFrame for Feature Relationship Analysis
```

```
correlation_df = df[
    [ "Population",
      "Participation_Rate",
      "Employment_Rate",
      "Unemployment_Rate",
      "Core_Housing_Need",
      "Bachelors_Degree_Pct",
      "Visible_Minority_Pct",
      "Immigrant_Population_Pct",
      "Renters_Pct",
      "Lone_Parent_Families_Pct",
      "Unaffordable_Housing",
      "Low_Income_AfterTax_Pct"
    ]].copy()

print(f"Correlation DataFrame Shape: {correlation_df.shape}")
display(correlation_df.head())
```

Correlation DataFrame Shape: (140, 12)

	Population	Participation_Rate	Employment_Rate	Unemployment_Rate	Core_Housing_Nee
0	29113	55.40	50.00	9.80	222
1	23757	59.00	53.20	9.80	179
2	12054	66.50	62.40	6.10	54
3	30526	70.60	65.80	6.70	242
4	27695	59.90	55.60	7.20	211

Section-9. Correlation Analysis

9.1 Correlation between Independent Variables and the Dependent Variable

Pearson Correlation Analysis

This section addresses **Research Question 1** by examining the linear relationships between the selected independent variables and the dependent variable, **Low_Income_AfterTax_Pct (LIA-Tax%)**, which is used as a proxy indicator of neighbourhood socioeconomic vulnerability.

Pearson correlation analysis is performed to quantify both the **strength** and **statistical significance** of the relationships between each independent variable and the dependent variable. The analysis is exploratory and is intended to identify statistical associations rather than establish causal relationships.

The independent variables were pre-specified during the data preparation stage (Notebook 1) based on the research objectives, the social vulnerability framework, and data availability.

Consequently, the correlation analysis is used to understand relationships among the selected variables rather than to perform feature selection, thereby reducing the risk of target-informed feature selection.

```
In [124... # Pearson Correlation Analysis with Statistical Significance

from scipy.stats import pearsonr

dependent_variable = "Low_Income_AfterTax_Pct"

results = []

for column in correlation_df.columns:
    if column == dependent_variable:
        continue

    r, p = pearsonr(
        correlation_df[column],
        correlation_df[dependent_variable])

    results.append({
        "Variable": column,
        "Pearson_r": round(r, 3),
        "P_value": p })

pearson_results = (
    pd.DataFrame(results)
    .sort_values(by="Pearson_r",
                 key=lambda s: s.abs(),
                 ascending=False))

pearson_results["Significance"] = pearson_results["P_value"].apply(
    lambda p: "p < 0.001" if p < 0.001 else f"p = {p:.3f}")

print("=" * 70)
print("PEARSON CORRELATION ANALYSIS")
print("=" * 70)

display(
    pearson_results[
        ["Variable", "Pearson_r", "Significance"]])

=====
PEARSON CORRELATION ANALYSIS
=====
```

	Variable	Pearson_r	Significance
10	Unaffordable_Housing	0.78	p < 0.001
3	Unemployment_Rate	0.73	p < 0.001
6	Visible_Minority_Pct	0.66	p < 0.001
7	Immigrant_Population_Pct	0.56	p < 0.001
4	Core_Housing_Need	0.55	p < 0.001
2	Employment_Rate	-0.47	p < 0.001
8	Renters_Pct	0.47	p < 0.001
9	Lone_Parent_Families_Pct	0.43	p < 0.001
1	Participation_Rate	-0.39	p < 0.001
5	Bachelors_Degree_Pct	-0.32	p < 0.001
0	Population	0.17	p = 0.045

Interpretation

The Pearson correlation analysis examined the linear relationships between the selected independent variables and the dependent variable, **Low_Income_AfterTax_Pct (LIA-Tax%)**, which is used as a proxy indicator of neighbourhood socioeconomic vulnerability.

The strongest positive associations with neighbourhood low-income prevalence were observed for **Unaffordable_Housing (r = 0.78)**, **Unemployment_Rate (r = 0.73)**, **Visible_Minority_Pct (r = 0.66)**, **Immigrant_Population_Pct (r = 0.56)**, and **Core_Housing_Need (r = 0.55)**. These results indicate that neighbourhoods experiencing higher housing affordability challenges, unemployment, larger visible minority and immigrant populations, and greater housing need generally exhibit higher levels of low-income prevalence.

Conversely, **Employment_Rate (r = -0.47)**, **Participation_Rate (r = -0.39)**, and **Bachelors_Degree_Pct (r = -0.32)** demonstrated negative correlations with LIA-Tax%, suggesting that neighbourhoods with stronger labour market participation and higher educational attainment are generally associated with lower neighbourhood-level low-income prevalence.

All independent variables were statistically significant at the 5% significance level. Most relationships were highly significant (**p < 0.001**), Population exhibited the weakest positive correlation (r = 0.17, p = 0.045), indicating that neighbourhood size alone has only a limited linear association with low-income prevalence. This finding further supports the decision to express count-based variables as population-normalized percentages, allowing the analysis to focus on socioeconomic characteristics rather than differences in neighbourhood size.

Overall, these findings provide empirical evidence supporting **Research Question 1** and establish the foundation for the multicollinearity assessment and dimensionality reduction analyses presented in the following sections.

9.2 Pearson Correlation Matrix

This section examines the Pearson correlation matrix to explore the linear relationships among the analytical variables included in the machine learning-ready dataset.

Unlike the previous analysis, which focused on the relationship between each independent variable and the dependent variable (LIA-Tax%), the correlation matrix evaluates relationships among all analytical variables. This provides an overview of the dataset's correlation structure, helps identify potential multicollinearity, and establishes the basis for the dimensionality reduction and neighbourhood segmentation analyses presented in Notebook 3.

Pearson correlation coefficients range from -1 to $+1$, where values closer to ± 1 indicate stronger linear relationships and values near 0 indicate weak or no linear association..

In [125...

```
# Pearson Correlation Matrix

pearson_corr = correlation_df.corr(method="pearson")

print("=" * 70)
print("PEARSON CORRELATION MATRIX")
print("=" * 70)

display(pearson_corr.round(3))
```

```
=====
PEARSON CORRELATION MATRIX
=====
```

	Population	Participation_Rate	Employment_Rate	Unemployment_Rate
Population	1.00	-0.01	-0.03	
Participation_Rate	-0.01	1.00	0.99	
Employment_Rate	-0.03	0.99	1.00	
Unemployment_Rate	0.13	-0.67	-0.77	
Core_Housing_Need	0.82	-0.12	-0.17	
Bachelors_Degree_Pct	0.10	0.62	0.65	
Visible_Minority_Pct	0.38	-0.66	-0.71	
Immigrant_Population_Pct	0.31	-0.72	-0.75	
Renters_Pct	0.06	0.45	0.39	
Lone_Parent_Families_Pct	-0.01	-0.56	-0.60	
Unaffordable_Housing	0.27	-0.06	-0.13	
Low_Income_AfterTax_Pct	0.17	-0.39	-0.47	

Interpretation

The Pearson correlation matrix provides an overview of the linear relationships among the analytical variables included in the machine learning-ready dataset. Several variables exhibit moderate to strong correlations, indicating that multiple socioeconomic indicators capture related neighbourhood characteristics.

The strongest positive relationship was observed between **Participation_Rate** and **Employment_Rate** ($r = 0.99$), suggesting that neighbourhoods with higher labour force participation generally also experience higher employment rates. Strong positive correlations were also observed between **Population** and **Core_Housing_Need** ($r = 0.82$), reflecting that larger neighbourhoods tend to contain more households experiencing core housing need.

Strong negative relationships were identified between **Employment_Rate** and **Unemployment_Rate** ($r = -0.77$), **Employment_Rate** and **Immigrant_Population_Pct** ($r = -0.75$), and **Participation_Rate** and **Immigrant_Population_Pct** ($r = -0.72$). These findings indicate that several labour market, demographic, and socioeconomic indicators are closely associated and may describe similar underlying neighbourhood characteristics.

The presence of these moderate to strong intercorrelations suggests the potential for multicollinearity among the independent variables. Consequently, a formal multicollinearity assessment is conducted in the following section, and the observed correlation structure also provides methodological justification for the Principal Component Analysis (PCA) performed

in Notebook 3 to identify the underlying dimensions of neighbourhood socioeconomic vulnerability.

9.3. Spearman Rank Correlation Analysis

Spearman rank correlation is performed as a complementary non-parametric analysis to evaluate the monotonic relationships between the selected independent variables and the dependent variable, **Low_Income_AfterTax_Pct (LIA-Tax%)**, which is used as a proxy indicator of neighbourhood socioeconomic vulnerability.

Unlike Pearson correlation, which measures linear relationships, Spearman correlation is based on ranked observations and is less sensitive to non-normality and outliers. Comparing the Pearson and Spearman results provides additional evidence regarding the robustness of the observed relationships.

As with the Pearson analysis, Spearman correlation is exploratory and is intended to identify statistical associations rather than establish causal relationships.

```
In [126... # Spearman Correlation Analysis with Statistical Significance

from scipy.stats import spearmanr

dependent_variable = "Low_Income_AfterTax_Pct"

results = []

for column in correlation_df.columns:
    if column == dependent_variable:
        continue

    rho, p = spearmanr(
        correlation_df[column],
        correlation_df[dependent_variable])

    results.append({
        "Variable": column,
        "Spearman_rho": round(rho, 3),
        "P_value": p })

spearman_results = (
    pd.DataFrame(results)
    .sort_values(by="Spearman_rho",
                 key=lambda s: s.abs(),
                 ascending=False))

spearman_results["Significance"] = spearman_results["P_value"].apply(
    lambda p: "p < 0.001" if p < 0.001 else f"p = {p:.3f}")

print("=" * 70)
print("SPEARMAN RANK CORRELATION ANALYSIS")
print("=" * 70)
```

```
display(
  spearman_results[
    ["Variable", "Spearman_rho", "Significance"]]
)
```

=====

SPEARMAN RANK CORRELATION ANALYSIS

=====

	Variable	Spearman_rho	Significance
3	Unemployment_Rate	0.76	p < 0.001
10	Unaffordable_Housing	0.75	p < 0.001
6	Visible_Minority_Pct	0.72	p < 0.001
4	Core_Housing_Need	0.65	p < 0.001
7	Immigrant_Population_Pct	0.61	p < 0.001
2	Employment_Rate	-0.52	p < 0.001
9	Lone_Parent_Families_Pct	0.50	p < 0.001
1	Participation_Rate	-0.45	p < 0.001
8	Renters_Pct	0.40	p < 0.001
5	Bachelors_Degree_Pct	-0.38	p < 0.001
0	Population	0.29	p < 0.001

Interpretation

The Spearman rank correlation analysis examined the monotonic relationships between the selected independent variables and the dependent variable, Low_Income_AfterTax_Pct (LIA-Tax%), which serves as a proxy measure of neighbourhood socioeconomic vulnerability. Unlike Pearson correlation, Spearman correlation evaluates relationships using the ranked order of observations rather than their raw values, making it less sensitive to outliers and departures from normality.

The strongest positive monotonic relationships were observed for Unemployment_Rate ($\rho = 0.76$), Unaffordable_Housing ($\rho = 0.75$), Visible_Minority_Pct ($\rho = 0.72$), Core_Housing_Need ($\rho = 0.65$), Immigrant_Population_Pct ($\rho = 0.61$), and Lone_Parent_Families_Pct ($\rho = 0.50$). These findings indicate that neighbourhoods experiencing higher unemployment, greater housing affordability challenges, larger immigrant and visible minority populations, and a higher proportion of lone-parent families generally exhibit higher levels of neighbourhood-level low-income prevalence.

The strongest negative monotonic relationships were observed for Employment_Rate ($\rho = -0.52$), Participation_Rate ($\rho = -0.45$), and Bachelor_Degree_Pct ($\rho = -0.38$). These results suggest that neighbourhoods with stronger labour market participation, higher employment

levels, and a greater proportion of residents holding bachelor's degrees generally experience lower neighbourhood-level low-income prevalence.

All relationships were statistically significant ($p < 0.001$), indicating strong evidence that the observed associations are unlikely to have occurred by random variation.

Overall, the Spearman correlation results closely mirror the findings from the Pearson correlation analysis, demonstrating that the identified relationships are consistent across both parametric (Pearson) and non-parametric (Spearman) correlation methods. This consistency indicates that the observed associations are robust regardless of whether they are evaluated using raw values or ranked observations, thereby strengthening the evidence supporting Research Question 1..

9.4 Spearman Correlation Matrix

This section presents the Spearman correlation matrix to examine the monotonic relationships among the analytical variables included in the machine learning-ready dataset.

Whereas the previous section evaluated the relationships between each independent variable and the dependent variable, the Spearman correlation matrix examines the relationships among all analytical variables using ranked observations. Comparing the Spearman and Pearson correlation matrices provides additional evidence regarding the consistency and robustness of the observed relationships while reducing sensitivity to non-normality and potential outliers.

The resulting correlation structure also provides supporting evidence for the multicollinearity assessment and the dimensionality reduction analysis performed in Notebook 3.

In [127...

```
# Spearman Correlation Matrix

spearman_corr = correlation_df.corr(method="spearman")

print("=" * 70)
print("SPEARMAN CORRELATION MATRIX")
print("=" * 70)

display(spearman_corr.round(3))
```

```
=====
SPEARMAN CORRELATION MATRIX
=====
```

	Population	Participation_Rate	Employment_Rate	Unemployment_Rate
Population	1.00	-0.14	-0.15	
Participation_Rate	-0.14	1.00	0.99	
Employment_Rate	-0.15	0.99	1.00	
Unemployment_Rate	0.23	-0.70	-0.78	
Core_Housing_Need	0.83	-0.20	-0.24	
Bachelors_Degree_Pct	0.00	0.57	0.61	
Visible_Minority_Pct	0.42	-0.68	-0.73	
Immigrant_Population_Pct	0.36	-0.75	-0.77	
Renters_Pct	0.05	0.40	0.36	
Lone_Parent_Families_Pct	0.05	-0.57	-0.61	
Unaffordable_Housing	0.33	-0.09	-0.14	
Low_Income_AfterTax_Pct	0.29	-0.45	-0.52	

9.5 Comparison of Pearson and Spearman Correlation Results

To evaluate the robustness of the correlation analysis, the Pearson and Spearman correlation coefficients were compared for each independent variable. Consistent results across both methods provide greater confidence that the observed relationships are stable and not substantially influenced by non-linearity or extreme observations.

In [128...

```
# Compare Pearson and Spearman Correlation Results

comparison = (
    pearson_results[["Variable", "Pearson_r"]].merge(spearman_results[["Variable",
comparison["Difference"] = (
    comparison["Pearson_r"] -
    comparison["Spearman_rho"])).round(3)

comparison = comparison.sort_values(
    by="Pearson_r",
    key=lambda x: x.abs(),
    ascending=False)

display(comparison.round(3))
```

	Variable	Pearson_r	Spearman_rho	Difference
0	Unaffordable_Housing	0.78	0.75	0.03
1	Unemployment_Rate	0.73	0.76	-0.02
2	Visible_Minority_Pct	0.66	0.72	-0.06
3	Immigrant_Population_Pct	0.56	0.61	-0.05
4	Core_Housing_Need	0.55	0.65	-0.10
5	Employment_Rate	-0.47	-0.52	0.04
6	Renters_Pct	0.47	0.40	0.07
7	Lone_Parent_Families_Pct	0.43	0.50	-0.07
8	Participation_Rate	-0.39	-0.45	0.06
9	Bachelors_Degree_Pct	-0.32	-0.38	0.06
10	Population	0.17	0.29	-0.12

Interpretation

The Pearson and Spearman correlation analyses produced highly consistent results for the majority of the socioeconomic indicators, demonstrating that the observed relationships with Low_Income_AfterTax_Pct, a proxy indicator of neighbourhood socioeconomic vulnerability, were robust across both correlation methods.

Differences between the Pearson and Spearman correlation coefficients were generally small (less than 0.10), indicating that the identified relationships were stable and not substantially influenced by non-linearity or extreme observations.

Across both methods, Unaffordable_Housing, Unemployment_Rate, Visible_Minority_Pct, and Immigrant_Population_Pct consistently showed the strongest positive associations with neighbourhood-level low-income prevalence. In contrast, Employment_Rate, Participation_Rate, and Bachelors_Degree_Pct consistently demonstrated strong negative associations. Although Core_Housing_Need exhibited a larger difference between the two methods, it remained an important housing-related socioeconomic indicator.

Overall, the close agreement between the Pearson and Spearman analyses provides strong evidence identifying the socioeconomic indicators most strongly associated with neighbourhood-level low-income prevalence, directly addressing Research Question 1 and supporting the subsequent multicollinearity assessment using the Variance Inflation Factor (VIF).

Section- 10. Correlation Heatmap

10.1. Pearson Correlation Heatmap

To complement the numerical correlation analyses, a correlation heatmap is generated to visualize the strength and direction of the relationships between the selected socioeconomic indicators.

The heatmap provides an intuitive overview of the correlation structure and assists in identifying groups of variables that exhibit similar relationships with neighbourhood-level low-income prevalence.

This visualization complements the Pearson and Spearman correlation analyses presented in the previous section.

```
In [129... # Pearson Correlation Heatmap

import matplotlib.pyplot as plt
import seaborn as sns

plt.figure(figsize=(14, 12))

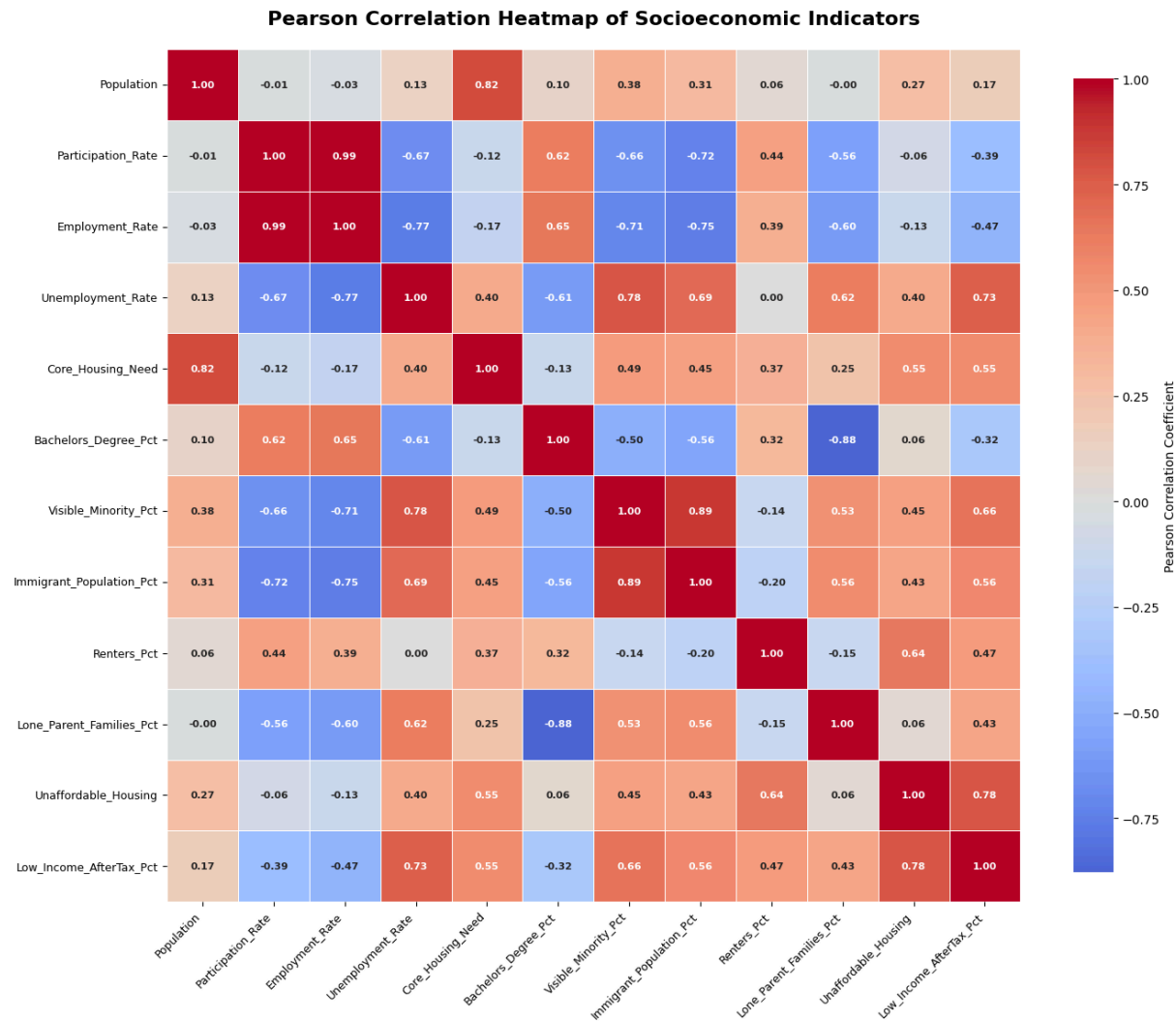
sns.heatmap(
    pearson_corr,
    annot=True,                # Show correlation coefficients
    fmt=".2f",                 # Display two decimal places
    cmap="coolwarm",          # Blue = negative, Red = positive
    center=0,                 # Center color scale at zero
    linewidths=0.6,
    linecolor="white",
    square=True,
    annot_kws={
        "size": 8,
        "weight": "bold"    },
    cbar_kws={
        "label": "Pearson Correlation Coefficient",
        "shrink": 0.85    })

plt.title(
    "Pearson Correlation Heatmap of Socioeconomic Indicators",
    fontsize=16,
    fontweight="bold",
    pad=20)

plt.xticks(rotation=45, ha="right", fontsize=9)
plt.yticks(rotation=0, fontsize=9)

plt.tight_layout()

plt.show()
```



Interpretation

The Pearson correlation heatmap provides a visual summary of the linear relationships among the selected socioeconomic indicators and Low_Income_AfterTax_Pct (LIM-AT). Housing-related variables, unemployment, and demographic indicators exhibit the strongest positive correlations with neighbourhood-level low-income prevalence, while employment, participation, and educational attainment show negative correlations.

The heatmap also identifies several strong correlations among the independent variables, suggesting potential multicollinearity. These relationships are examined further in the subsequent Variance Inflation Factor (VIF) analysis.

10.2 Spearman Correlation Heatmap

```
In [130... # Spearman correlation matrix
spearman_corr = correlation_df.corr(method="spearman")

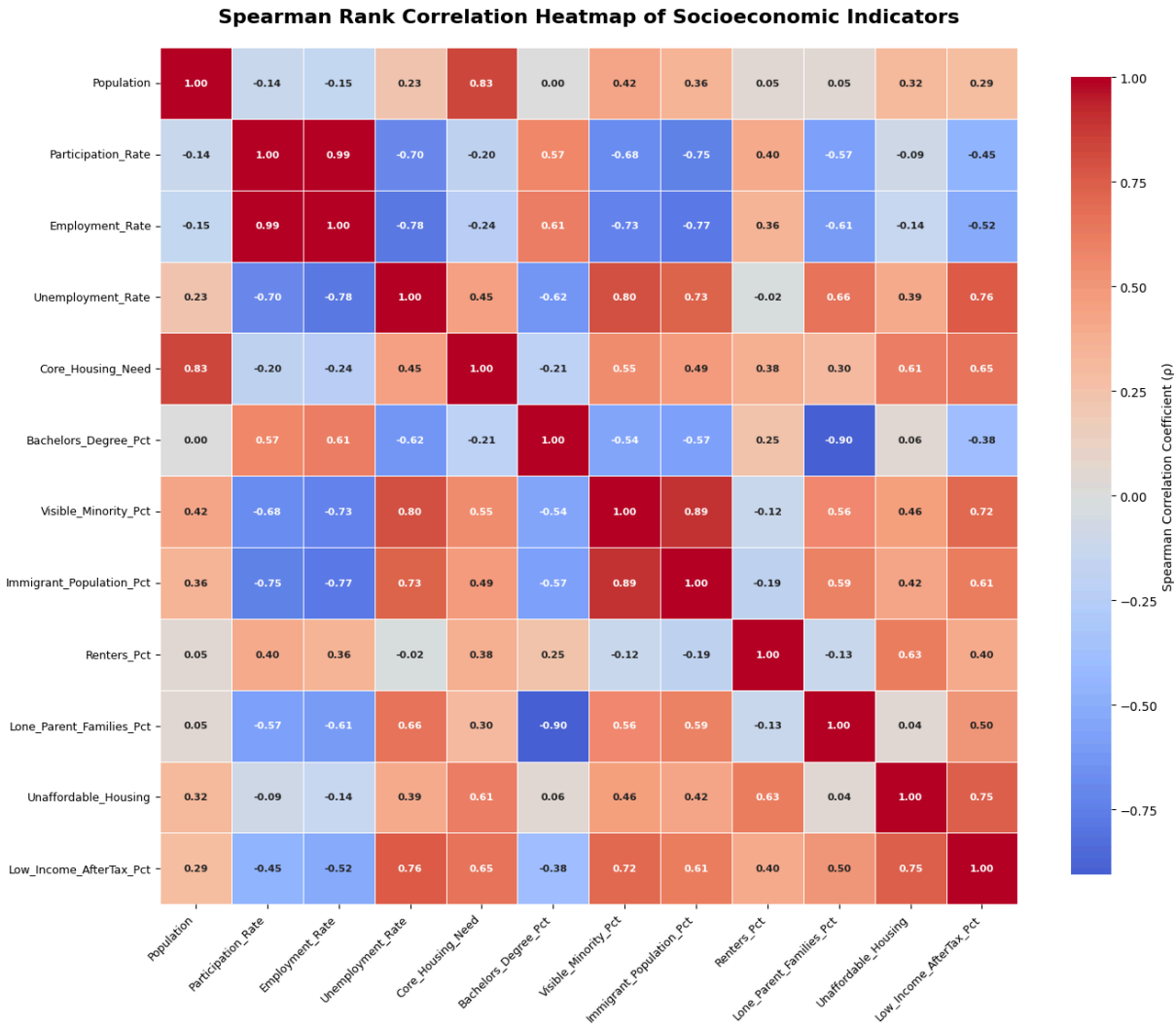
# Plot Spearman correlation heatmap
plt.figure(figsize=(14, 12))
```

```
sns.heatmap(
    spearman_corr,
    annot=True,
    fmt=".2f",
    cmap="coolwarm",
    center=0,
    linewidths=0.6,
    linecolor="white",
    square=True,
    annot_kws={
        "size": 8,
        "weight": "bold"    },
    cbar_kws={
        "label": "Spearman Correlation Coefficient ( $\rho$ )",
        "shrink": 0.85    })

plt.title(
    "Spearman Rank Correlation Heatmap of Socioeconomic Indicators",
    fontsize=16,
    fontweight="bold",
    pad=20)

plt.xticks(rotation=45, ha="right", fontsize=9)
plt.yticks(rotation=0, fontsize=9)

plt.tight_layout()
plt.show()
```



Interpretation

The Spearman rank correlation heatmap provides a visual summary of the monotonic relationships among the selected socioeconomic indicators and **Low_Income_AfterTax_Pct (LIA-Tax%)**, which serves as a proxy measure of neighbourhood socioeconomic vulnerability.

The heatmap confirms the findings of the Spearman correlation analysis, with **Unemployment_Rate**, **Unaffordable_Housing**, **Visible_Minority_Pct**, **Core_Housing_Need**, and **Immigrant_Population_Pct** showing strong positive relationships with neighbourhood-level low-income prevalence. In contrast, **Employment_Rate**, **Participation_Rate**, and **Bachelors_Degree_Pct** exhibit negative relationships.

The heatmap also reveals several strong correlations among the predictor variables, indicating potential multicollinearity. Overall, the Spearman heatmap closely resembles the Pearson heatmap, demonstrating that the observed relationships are robust across both parametric and non-parametric correlation methods. These findings support the subsequent Variance Inflation Factor (VIF) analysis to assess multicollinearity among the predictors.

Summary of Correlation Analyses

The Pearson and Spearman correlation analyses produced highly consistent rankings of the selected socioeconomic indicators. The small differences observed between the two correlation coefficients indicate that the identified relationships are robust and not sensitive to the choice of correlation method.

Across both analyses, **Unaffordable_Housing**, **Unemployment_Rate**, **Visible_Minority_Pct**, and **Immigrant_Population_Pct** consistently emerged as the strongest positive predictors of neighbourhood-level low-income prevalence. Similarly, **Employment_Rate**, **Participation_Rate**, and **Bachelors_Degree_Pct** consistently demonstrated negative associations.

The agreement between the two correlation methods provides strong statistical evidence that these socioeconomic indicators are meaningfully associated with neighbourhood vulnerability and low-income prevalence, thereby directly addressing Research Question 1.

11. Multicollinearity Assessment (Variance Inflation Factor)

Q: Which predictors are too similar to each other?

The previous correlation analyses identified socioeconomic indicators that are significantly associated with the dependent variable, Low_Income_AfterTax_Pct (LIM-AT). However, independent variables may also be highly correlated with one another, resulting in multicollinearity.

Multicollinearity can reduce the stability and interpretability of regression models by introducing redundant information and inflating coefficient estimates.

To assess the degree of multicollinearity among the selected independent variables, the Variance Inflation Factor (VIF) is calculated for each variable. Variables with high VIF values indicate potential redundancy and should be interpreted carefully before proceeding with regression and predictive modelling.

In [131...

```
from statsmodels.stats.outliers_influence import variance_inflation_factor
from statsmodels.tools.tools import add_constant

X = correlation_df.drop(columns=[
    "Population",
    "Low_Income_AfterTax_Pct"])

X = add_constant(X)

vif = pd.DataFrame({
    "Variable": X.columns,
    "VIF": [
        variance_inflation_factor(X.values, i)
```

```
for i in range(X.shape[1])    ]})

vif = (
    vif[vif["Variable"] != "const"]
    .sort_values(by="VIF", ascending=False))

display(vif.round(2))
```

	Variable	VIF
2	Employment_Rate	3013.66
1	Participation_Rate	2280.63
3	Unemployment_Rate	108.01
7	Immigrant_Population_Pct	7.51
6	Visible_Minority_Pct	7.35
5	Bachelors_Degree_Pct	5.88
9	Lone_Parent_Families_Pct	5.23
10	Unaffordable_Housing	4.43
8	Renters_Pct	4.25
4	Core_Housing_Need	1.88

Interpretation

The Variance Inflation Factor (VIF) analysis was conducted to assess multicollinearity among the selected independent socioeconomic variables. The results indicate that **Employment_Rate**, **Participation_Rate**, and **Unemployment_Rate** exhibit very high VIF values, reflecting the strong relationships that naturally exist among these labour market indicators. Several additional variables, including **Bachelors_Degree_Pct**, **Visible_Minority_Pct**, **Immigrant_Population_Pct**, and **Lone_Parent_Families_Pct**, demonstrate moderate levels of multicollinearity, while **Core_Housing_Need**, **Renters_Pct**, and **Unaffordable_Housing** remain within acceptable ranges.

These findings suggest that some predictors contain overlapping information. However, multicollinearity alone does not justify removing variables, particularly when they represent distinct socioeconomic dimensions that are important to the research objectives. Instead, the VIF results provide guidance for the subsequent feature selection stage, where statistical evidence and domain knowledge will be considered together when selecting variables for the final predictive model.

12. Feature Selection Strategy

Feature selection is a critical step in predictive modelling because it identifies the subset of variables that provides the most useful information for explaining the response variable while reducing redundancy and multicollinearity.

Several feature selection approaches are commonly used in regression modelling:

- **Forward Selection** begins with no predictors and sequentially adds variables that produce the greatest improvement in model performance.
- **Backward Elimination** begins with all candidate predictors and iteratively removes variables that contribute the least to the model.
- **Stepwise Regression** combines both approaches by allowing variables to be added or removed throughout the selection process.

Although these automated procedures are widely used, this study adopted a research-driven feature selection strategy. Predictor variables were selected based on prior socioeconomic literature, exploratory data analysis, Pearson and Spearman correlation analyses, and Variance Inflation Factor (VIF) assessment.

This approach ensured that the final predictor set remained statistically appropriate while preserving variables with established theoretical importance for explaining neighbourhood socioeconomic vulnerability.

In [132...

```
import pandas as pd
import numpy as np

X = correlation_df.drop(columns=["Low_Income_AfterTax_Pct"])
y = correlation_df["Low_Income_AfterTax_Pct"]

print(X.columns.tolist())
```

```
['Population', 'Participation_Rate', 'Employment_Rate', 'Unemployment_Rate', 'Core_Housing_Need', 'Bachelors_Degree_Pct', 'Visible_Minority_Pct', 'Immigrant_Population_Pct', 'Renters_Pct', 'Lone_Parent_Families_Pct', 'Unaffordable_Housing']
```

In [133...

```
import statsmodels.api as sm
```

In [134...

```
def forward_selection(X, y, significance_level=0.05):

    initial_features = []
    remaining_features = list(X.columns)

    while remaining_features:

        pvals = pd.Series(dtype=float)

        for feature in remaining_features:

            model = sm.OLS(
                y,
                sm.add_constant(
                    X[initial_features + [feature]]
```

```

        )
    ).fit()

    pvals.loc[feature] = model.pvalues[feature]

    min_p = pvals.min()

    if min_p < significance_level:

        best_feature = pvals.idxmin()

        initial_features.append(best_feature)

        remaining_features.remove(best_feature)

    else:
        break

    return initial_features

forward_features = forward_selection(X, y)

print("Forward Selection:")
print(forward_features)

```

Forward Selection:

['Unaffordable_Housing', 'Unemployment_Rate', 'Renters_Pct', 'Participation_Rate', 'Visible_Minority_Pct', 'Immigrant_Population_Pct', 'Lone_Parent_Families_Pct', 'Employment_Rate']

In [135...

```

def backward_elimination(X, y, significance_level=0.05):

    features = list(X.columns)

    while True:

        model = sm.OLS(
            y,
            sm.add_constant(X[features])
        ).fit()

        pvalues = model.pvalues.iloc[1:]

        max_p = pvalues.max()

        if max_p > significance_level:

            worst_feature = pvalues.idxmax()

            features.remove(worst_feature)

        else:
            break

    return features

```

```
backward_features = backward_elimination(X, y)

print("Backward Elimination:")
print(backward_features)
```

Backward Elimination:

```
['Population', 'Participation_Rate', 'Employment_Rate', 'Unemployment_Rate', 'Core_Housing_Need', 'Visible_Minority_Pct', 'Immigrant_Population_Pct', 'Renters_Pct', 'Unaffordable_Housing']
```

```
In [136... def stepwise_selection(X, y,
                      threshold_in=0.05,
                      threshold_out=0.05):

    included = []

    while True:

        changed = False

        excluded = list(set(X.columns) - set(included))

        new_pvals = pd.Series(dtype=float)

        for feature in excluded:

            model = sm.OLS(
                y,
                sm.add_constant(
                    X[included + [feature]]
                )
            ).fit()

            new_pvals.loc[feature] = model.pvalues[feature]

        if not new_pvals.empty:

            best_p = new_pvals.min()

            if best_p < threshold_in:

                best_feature = new_pvals.idxmin()

                included.append(best_feature)

                changed = True

        if included:

            model = sm.OLS(
                y,
                sm.add_constant(X[included])
            ).fit()

            pvalues = model.pvalues.iloc[1:]
```

```

        worst_p = pvalues.max()

        if worst_p > threshold_out:

            worst_feature = pvalues.idxmax()

            included.remove(worst_feature)

            changed = True

        if not changed:
            break

    return included

stepwise_features = stepwise_selection(X, y)

print("Stepwise Selection:")
print(stepwise_features)

```

Stepwise Selection:

```

['Unaffordable_Housing', 'Unemployment_Rate', 'Renters_Pct', 'Participation_Rate',
'Visible_Minority_Pct', 'Immigrant_Population_Pct', 'Lone_Parent_Families_Pct', 'Emp
loyment_Rate']

```

```

In [137... import statsmodels.api as sm
def evaluate_model(features, X, y):

    X_model = sm.add_constant(X[features])

    model = sm.OLS(y, X_model).fit()

    rmse = np.sqrt(np.mean(model.resid**2))

    return {
        "Variables": len(features),
        "Adjusted R²": model.rsquared_adj,
        "AIC": model.aic,
        "BIC": model.bic,
        "RMSE": rmse    }

```

```

In [138... forward_results = evaluate_model(forward_features, X, y)

backward_results = evaluate_model(backward_features, X, y)

stepwise_results = evaluate_model(stepwise_features, X, y)

```

```

In [139... comparison = pd.DataFrame({
    "Forward": forward_results,
    "Backward": backward_results,
    "Stepwise": stepwise_results
}).T

comparison

```

Out[139...

	Variables	Adjusted R ²	AIC	BIC	RMSE
Forward	8.00	0.88	689.54	716.01	2.66
Backward	9.00	0.89	679.11	708.53	2.55
Stepwise	8.00	0.88	689.54	716.01	2.66

In [140...

```
backward_model = sm.OLS(y, sm.add_constant(X[backward_features])).fit()  
  
print(backward_model.summary())
```

OLS Regression Results

```

=====
Dep. Variable:      Low_Income_AfterTax_Pct      R-squared:                0.895
Model:              OLS                        Adj. R-squared:           0.888
Method:             Least Squares              F-statistic:             123.2
Date:               Tue, 21 Jul 2026            Prob (F-statistic):       3.40e-59
Time:               12:19:34                    Log-Likelihood:          -329.55
No. Observations:   140                        AIC:                     679.1
Df Residuals:       130                        BIC:                     708.5
Df Model:           9
Covariance Type:    nonrobust
=====

```

```

=====
                                coef      std err          t      P>|t|      [0.025
0.975]
-----
const                -6.5690      10.736      -0.612      0.542     -27.809
14.671
Population            -0.0003     5.66e-05     -4.673      0.000     -0.000
-0.000
Participation_Rate    -4.1735       1.740     -2.398      0.018     -7.616
-0.731
Employment_Rate       4.1986       1.875       2.239      0.027       0.489
7.908
Unemployment_Rate     3.5053       1.228       2.856      0.005       1.077
5.934
Core_Housing_Need     0.0026       0.001       4.015      0.000       0.001
0.004
Visible_Minority_Pct  0.1560       0.029       5.392      0.000       0.099
0.213
Immigrant_Population_Pct -0.1653     0.044     -3.720      0.000     -0.253
-0.077
Renters_Pct           0.2203       0.056       3.949      0.000       0.110
0.331
Unaffordable_Housing  0.3993       0.063       6.310      0.000       0.274
0.524
=====

```

```

=====
Omnibus:              29.104      Durbin-Watson:           2.102
Prob(Omnibus):        0.000      Jarque-Bera (JB):        62.563
Skew:                 0.874      Prob(JB):                2.60e-14
Kurtosis:             5.769      Cond. No.                1.08e+06
=====

```

Notes:

- [1] Standard Errors assume that the covariance matrix of the errors is correctly specified.
- [2] The condition number is large, 1.08e+06. This might indicate that there are strong multicollinearity or other numerical problems.

Interpration

Method	Variables	Adjusted R ²	AIC	BIC	RMSE
Forward Selection	8	0.88	689.54	716.01	2.66

Method	Variables	Adjusted R ²	AIC	BIC	RMSE
Backward Elimination	9	0.89	679.11	708.53	2.55
Stepwise Selection	8	0.88	689.54	716.01	2.66

Forward Selection, Backward Elimination, and Stepwise Selection were applied to the 12 candidate socioeconomic variables to identify predictors that contribute most to explaining neighbourhood-level low-income prevalence. Forward Selection and Stepwise Selection selected the same eight predictors, whereas Backward Elimination retained nine predictors.

As shown in Table, the Backward Elimination model achieved the highest Adjusted R² (0.89), the lowest RMSE (2.55), and the lowest AIC (679.11) and BIC (708.53), indicating the strongest overall statistical performance among the three methods. However, the regression diagnostics also confirmed substantial multicollinearity, consistent with the high VIF values identified earlier. Therefore, the feature selection results should be interpreted together with the VIF analysis rather than based solely on model fit.

Overall, the three automatic feature selection methods produced largely consistent predictor sets, reinforcing the importance of housing affordability, labour market conditions, and demographic characteristics in explaining neighbourhood-level low-income prevalence

13. Research Question 1 Summary

Research Question 1

Which socioeconomic variables are the strongest predictors of neighbourhood-level low-income prevalence in Toronto?

Quantitative Findings

The statistical analyses consistently identified several socioeconomic indicators as the strongest predictors of neighbourhood-level low-income prevalence.

Using Pearson correlation analysis, the strongest positive linear associations with Low_Income_AfterTax_Pct were:

Predictor	Pearson (r)	Spearman (ρ)
Unaffordable_Housing	0.78	0.74
Unemployment_Rate	0.73	0.76
Visible_Minority_Pct	0.66	0.72
Immigrant_Population_Pct	0.56	0.61

The strongest negative associations were:

Predictor	Pearson (r)	Spearman (p)
Employment_Rate	-0.47	-0.52
Participation_Rate	-0.39	-0.45
Bachelors_Degree_Pct	-0.32	-0.38

All reported correlations were statistically significant ($p < 0.001$). The Pearson and Spearman analyses produced very similar results, indicating that the observed relationships were robust across both parametric and non-parametric correlation methods.

The VIF analysis identified substantial multicollinearity among Employment_Rate, Participation_Rate, and Unemployment_Rate, indicating that these labour market indicators contain overlapping information. The remaining predictors demonstrated acceptable levels of multicollinearity.

To further evaluate the candidate variables, Forward Selection, Backward Elimination, and Stepwise Selection were applied using OLS regression. These methods assessed which combination of the twelve candidate socioeconomic variables produced the strongest predictive model.

Automatic Feature Selection Results

Method	Variables Selected	Adjusted R ²	AIC	BIC	RMSE
Forward Selection	8	0.88	689.54	716.01	2.66
Backward Elimination	9	0.89	679.11	708.53	2.55
Stepwise Selection	8	0.88	689.54	716.01	2.66

Forward Selection and Stepwise Selection identified the same eight predictors, while Backward Elimination retained one additional predictor and achieved the strongest statistical performance, with the highest Adjusted R² and the lowest AIC, BIC, and RMSE values. Although the regression diagnostics indicated continued multicollinearity among several labour market variables, the three automatic feature selection methods produced largely consistent predictor sets.

Overall Conclusion

Overall, the correlation analyses, VIF assessment, and automatic feature selection methods produced consistent findings. Housing affordability, unemployment, employment and participation rates, visible minority population, immigrant population, renters, and core housing need emerged as the most influential socioeconomic indicators associated with neighbourhood-level low-income prevalence in Toronto. The agreement between the theory-driven variable selection and the automatic feature selection

methods provides additional confidence that the selected predictors are appropriate for subsequent predictive modelling.

14. Conclusion

This notebook examined the statistical relationships between selected socioeconomic indicators and the dependent variable, Low_Income_AfterTax_Pct (LIM-AT), which serves in this study as a proxy measure of neighbourhood socioeconomic vulnerability. The analyses were conducted using the machine learning-ready dataset developed in Notebook 1.

Using descriptive statistics, Pearson and Spearman correlation analyses, statistical significance testing, correlation heatmaps, Variance Inflation Factor (VIF) analysis, and automatic feature selection methods, the study identified Unaffordable_Housing, Unemployment_Rate, Visible_Minority_Pct, Immigrant_Population_Pct, Employment_Rate, Participation_Rate, Renters_Pct, and Core_Housing_Need as the socioeconomic indicators most strongly associated with neighbourhood-level low-income prevalence. Pearson and Spearman analyses produced highly consistent results, demonstrating that these relationships were robust across both parametric and non-parametric methods.

The VIF analysis identified substantial multicollinearity among the labour market indicators, while the automatic feature selection methods produced largely consistent predictor sets. Although Backward Elimination achieved the strongest statistical performance, the agreement among the three feature selection methods provides additional validation of the theory-driven candidate variables selected in Notebook 1.

Overall, these findings provide a statistically supported answer to Research Question 1 by identifying the socioeconomic indicators most strongly associated with neighbourhood-level low-income prevalence in Toronto. The validated predictor set establishes a strong foundation for the subsequent dimensionality reduction and predictive modelling analyses presented in the following notebooks.

15. Export Statistical Analysis Results

Purpose

Following the completion of the statistical analyses, the key analytical outputs are exported to a consolidated Excel workbook to support reproducibility, documentation, and future modelling activities.

The exported workbook provides a structured summary of the quantitative findings generated throughout this notebook, including descriptive statistics, correlation analyses, comparison of correlation methods, and multicollinearity assessment.

The workbook serves as a supplementary analytical report that documents the statistical evidence used to answer **Research Question 1** and provides a reference for subsequent

notebooks in this capstone project.

Output

The following worksheets are exported to a single Excel workbook:

- **RQ1 Summary** – Quantitative summary answering Research Question 1.
- **Descriptive Statistics** – Summary statistics for all selected variables.
- **Pearson Correlation** – Pearson correlation coefficients, significance levels, and statistical results.
- **Spearman Correlation** – Spearman rank correlation coefficients, significance levels, and statistical results.
- **Correlation Comparison** – Comparison between Pearson and Spearman correlation coefficients.
- **VIF Results** – Variance Inflation Factor (VIF) values and multicollinearity assessment.

The workbook is saved in the **reports** directory as:

02_Statistical_Analysis_Results.xlsx

```
In [141... rq1_summary = pd.DataFrame({
    "Research Question": [
        "Which socioeconomic variables are the strongest predictors of neighbourhoo
    "Strongest Positive Predictor": [
        "Unaffordable_Housing (Pearson r = 0.78, Spearman p = 0.74)"],
    "Second Strongest Predictor": [
        "Unemployment_Rate (Pearson r = 0.73, Spearman p = 0.76)"],
    "Third Strongest Predictor": [
        "Visible_Minority_Pct (Pearson r = 0.66, Spearman p = 0.72)"],
    "All Significant": ["Yes (p < 0.001)"]})
```

```
In [144... with pd.ExcelWriter(excel_path, engine="openpyxl") as writer:

    rq1_summary.to_excel(
        writer,
        sheet_name="RQ1_Summary",
        index=False    )

    df.describe().round(3).to_excel(
        writer,
        sheet_name="Descriptive_Statistics"    )

    pearson_results.round(3).to_excel(
        writer,
        sheet_name="Pearson_Correlation",
        index=False    )

    spearman_results.round(3).to_excel(
        writer,
        sheet_name="Spearman_Correlation",
        index=False    )
```

```

vif.round(3).to_excel(
    writer,
    sheet_name="VIF_Results",
    index=False
)

feature_selection_methods.to_excel(
    writer,
    sheet_name="Feature_Selection",
    index=False
)

comparison.round(3).to_excel(
    writer,
    sheet_name="Model_Performance"
)

print("="*60)
print("EXPORT COMPLETED")
print("="*60)
print(excel_path)

```

```

=====
EXPORT COMPLETED
=====
c:\Users\ksalyamova\Desktop\CIND 820\CIND820-Capstone-Project\reports\02_Statistical
_Analysis_Results.xlsx

```

```

In [145... analysis_ready = df.copy()

analysis_ready.to_csv(
    project_root /
    "data" /
    "processed" /
    "neighbourhood_profiles_2016_analysis_ready.csv",
    index=False)

print("Analysis-ready dataset exported successfully.")

```

Analysis-ready dataset exported successfully.